

Towards Imperceptible Color JPEG Image Hiding: Multi-Range Representations-Driven Adversarial Stego Generation

Junxue Yang, Xin Liao*, Weixuan Tang, Jianhua Yang, Zhen Xia*, Zheng Qin

Abstract—Image hiding fully explores the hidden potential of deep learning-based models, aiming to conceal image-level messages within cover images and reveal them from stego images to achieve covert communication. Although image hiding has made significant progress, achieving imperceptible color JPEG image hiding remains challenging. Existing hiding schemes are easily detected by the naked eyes or steganalyzers due to the cover type confined to the spatial domain, single-range feature extraction and attacks, and insufficient loss constraints. In this paper, we propose a multi-range representations-driven adversarial stego generation framework called MRAG for color JPEG image hiding. The proposed MRAG can address the aforementioned shortcomings within a unified framework, enabling steganalysis imperceptibility in color JPEG image hiding while ensuring secret restorability and visual imperceptibility. Specifically, MRAG integrates the local-range characteristic of the convolution and the global-range modeling of the transformer. This design stems from the fact that steganalyzers typically combine local-range and global-range information to better capture hidden traces. Meanwhile, we introduce a loss function, termed the features angle-norm disentanglement loss, to launch multi-range representations-driven feature-level adversarial attacks. It computes the adversarial loss between covers and stegos based on the surrogate steganalyzer's classified features, i.e., the features before the last fully connected layer. Under the dual constraints of features angle and norm, MRAG can delicately encode the concatenation of cover and secret into subtle adversarial perturbations from local and global ranges relevant to steganalysis. Therefore, the resulting stego can achieve visual and steganalysis imperceptibility. Moreover, coarse-grained and fine-grained frequency decomposition operations are devised to transform the input, introducing multi-grained frequency information. Extensive experiments demonstrate that MRAG can achieve state-of-the-art performance.

Index Terms—color JPEG image hiding, multi-range adversarial attacks, features angle-norm disentanglement.

I. INTRODUCTION

IMAGE steganography is the art and science of sheltering confidential information in publicly available cover images

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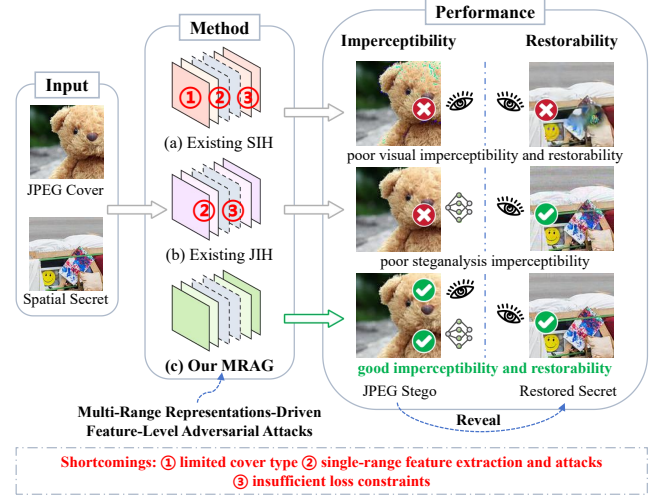


Fig. 1. The illustration shows the performance distinctions between our MRAG and existing image hiding methods when the cover is a color JPEG image. (a) Existing SIH method, due to the limited cover type, single-range feature extraction and attacks, and insufficient loss constraints, produces visible artifacts in the JPEG stego and the restored secret, resulting in poor visual imperceptibility and restorability. (b) Existing JIH method has poor steganalysis imperceptibility due to single-range feature extraction and insufficient loss constraints. (c) Our MRAG, which combines multi-range representations-driven feature-level adversarial attacks, has satisfying secret restorability, visual imperceptibility, and steganalysis imperceptibility.

to obtain stego ones. The resulting stegos should be as unsuspecting as possible, from which secret can be revealed [1]. It typically has three primary metrics [2]: the capacity regarding embedded payloads, the visual imperceptibility by the naked eyes, and the steganalysis imperceptibility against the steganalyzers. Particularly, robustness is a focused metric of robust steganography against the lossy channel. Traditional steganographic methods [3]–[5] and some deep learning-based methods [6]–[8] maintain the bit-level payloads, typically below 0.4 bits per pixel (BPP). The traditional can reach 1 BPP at most, while the deep learning-based can go as high as 4.5 BPP [9]. In light of this, Baluja [10] fully explores the potential of deep learning-based steganographic models, successfully embedding a spatial RGB image into another of the same size, known as image hiding. Three natural advantages of image hiding are also pointed out: 1) Since the secret is an image with high inherent correlations, the difficulty of network learning can be reduced. 2) The secret itself has a certain fault tolerance. 3) The payload can reach 24 BPP with a 1:1 ratio between the secret and cover. The success of this pioneering work, along with three natural advantages,

facilitates the flourishing development of image hiding.

The act of hiding yields embedded artifacts inevitably. Whether artifacts are discovered usually depends on three factors [11]: the hidden capacity, the imperceptibility of the algorithm, and the texture of the image itself. The likelihood of detection is negatively correlated with the latter two factors, but positively correlated with the capacity. In this sense, the capacity advantage of image hiding also makes it more vulnerable to detection, turning into a disadvantage. Most hiding methods [9], [12]–[17] default to such a disadvantage, preferring to improve capacity, visual imperceptibility, and robustness. Only several works [18]–[20] explore the steganalysis imperceptibility utilizing generative adversarial networks (GAN) or adversarial attacks. Due to the inherent instability and convergence difficulties in min-max optimization, GAN-based approaches are usually poorer [21]. To further improve steganalysis imperceptibility, adversarial attacks-based methods typically employ a pre-trained steganalyzer and introduce an adversarial loss based on the surrogate steganalyzer's output. Nevertheless, the cover type of the aforementioned methods is exclusively lossless spatial RGB images. We term these methods as spatial image hiding (SIH) for brevity.

When the cover is a color JPEG image, the ubiquitous lossy image format, SIH methods inherently perform poorly due to the limitation of the cover type. Compared to SIH, JPEG image hiding (JIH) poses unique challenges due to lossy decompression/compression, JPEG-related frequency information, and compressed embedding space. Specifically, for SIH methods, it is necessary to undergo decompression (for the hiding network), compression (from spatial stego image to JPEG stego image), and decompression (for the revealing network). The cumulative information loss caused by decompression/compression may result in visible color distortion, blocking, blurring, and banding artifacts in the generated stego and recovered secret image. Furthermore, SIH methods operate in the pixel domain, where image statistics are relatively continuous and stable. However, JPEG images are represented in the frequency domain, with block-wise discrete cosine transform (DCT) introducing discontinuities and statistical dependencies across DCT coefficients. Embedding strategies optimized for spatial images become mismatched with the characteristics of JPEG images. In addition, JPEG images are already compressed, leaving less redundancy available for information embedding compared to spatial images. This makes it more difficult to achieve image-level hiding while maintaining imperceptibility. Even the robust SIH method that accounts for channel noise, such as JPEG compression, still treats JPEG images as spatial inputs, resulting in unsatisfactory performance, as shown in Fig.1 (a).

Additionally, existing SIH methods are restricted to single-domain feature extraction and attacks, lacking steganalysis domain knowledge. The adversarial loss is typically computed as cross-entropy loss based on the surrogate steganalyzer's output. The recent EFDR [22] is currently the only work that uses color JPEG images as the covers. EFDR only evaluates steganalysis imperceptibility experimentally and does not consider it in the model design. Therefore, although EFDR can effectively mitigate visible artifacts, its steganalysis

imperceptibility remains unsatisfactory due to single-range feature extraction and insufficient loss constraints, as depicted in Fig.1 (b). In summary, although image hiding has made significant progress, achieving imperceptible color JPEG image hiding remains challenging. Existing hiding schemes are easily detected by the naked eyes or steganalyzers due to the cover type confined to the spatial domain, single-range feature extraction and attacks, and insufficient loss constraints. In this paper, we attempt to develop a novel adversarial stego generation framework for color JPEG image hiding, enabling steganalysis imperceptibility while ensuring secret restorability and visual imperceptibility, as illustrated in Fig.1 (c).

Building on the aforementioned analysis and summary, we construct a multi-range representations-driven adversarial stego generation framework called MRAG. The proposed MRAG can address the aforementioned shortcomings within a unified framework, enabling steganalysis imperceptibility in color JPEG image hiding while ensuring secret restorability and visual imperceptibility. Considering that steganalyzers capture subtle changes using both local-range and global-range information [23], MRAG integrates the local-range characteristic of the convolution and the global-range modeling of the transformer, introducing multi-range representations from a steganalysis perspective. Specifically, MRAG comprises four key components: the invertible local-range branch, the invertible global-range branch, the adaptive fusion module, and the surrogate steganalyzer. A loss function, termed the features angle-norm disentanglement loss, is incorporated to launch multi-range representations-driven feature-level adversarial attacks. It computes the adversarial loss between covers and generated stegos based on the surrogate steganalyzer's classified features. Therefore, MRAG is constrained to delicately encode the concatenation of cover and secret into subtle adversarial perturbations from local and global ranges relevant to steganalysis. We also tailor the coarse-grained frequency decomposition (CFD) and the fine-grained frequency decomposition (FFD) to transform the input. According to the shape of the obtained transformed images, the transformed images are used as the input for the local and global branches, respectively, naturally introducing multi-grained frequency information. Our contributions can be summarized as follows:

- We propose a multi-range representations-driven adversarial stego generation framework called MRAG for color JPEG image hiding, enabling steganalysis imperceptibility while ensuring secret restorability and visual imperceptibility. To the best of our knowledge, this is the first attempt to explore steganalysis imperceptibility in color JPEG image hiding, advancing toward more imperceptible color JPEG image hiding.
- We introduce a loss based on the surrogate steganalyzer's classified features, termed the features angle-norm disentanglement loss, which is incorporated to launch multi-range representations-driven feature-level adversarial attacks. Under the dual constraints of features angle and norm, MRAG can delicately encode the concatenation of cover and secret into subtle adversarial perturbations from local and global ranges relevant to steganalysis.

TABLE I
SUMMARIES OF EXISTING IMAGE HIDING METHODS FROM THREE DIMENSIONS: COVER TYPE, MODEL ARCHITECTURE, AND PERFORMANCE METRICS.

Cover Type	Hiding Method	Model Architecture	Performance Metrics			
			Visual Imperceptibility	Steganalysis Imperceptibility	Capacity	Robustness
Spatial	Baluja17 [10]	vanilla convolution	L2 loss	/	single	/
	Baluja20 [11]	vanilla convolution	L2 loss	/	multiple	/
	UDH [12]	cover-agnostic, simplified U-Net	L2 loss	/	single	/
	ISN [9]	INN, DenseNet-based affine coupling	L2, L1 losses	/	single	/
	HiNet [13]	INN, DenseNet-based affine coupling, DWT	L2, wavelet losses	/	single	/
	DeepMIH [14]	INN, importance map module, DWT, enhanced affine coupling	L2, wavelet, perceptual, importance, stage losses	/	multiple	/
	DenseJIN [15]	INN, noninvertible Unet-based module, DWT, densely enhanced affine coupling	L2, wavelet losses	/	single	/
	EFCA-DIH [16]	INN, Sobel operator, DWT, coordinate attention-based affine coupling	L2, wavelet, edge losses	/	single	/
	CAIS [18]	GAN, global and part discriminators	L1, perceptual losses	discriminators' output -based cross-entropy loss	single	/
	USAP [19]	adversarial attacks, pre-trained classifier	L2 loss	classifier's output -based cross-entropy loss	single	/
JPEG	DIH-OAIN [20]	adversarial attacks, pre-trained steganalyzer, INN, dual-resolution branches-based and priori knowledge-guided affine coupling	L2 loss	steganalyzer's output -based variant cross-entropy	single	/
	RIIS [17]	conditional INN, distortion layers, distortion-guided affine coupling	L2, fourier losses	/	single	distortion-guided, robust training
	EFDR [22]	INN, transformer-based affine coupling, fine-grained DCT representations	L2 loss	/	single	/
	MRAG (Proposed)	multi-range adversarial attacks, features angle-norm disentanglement, multi-grained frequency information, INN	L2 loss	steganalyzer's classified features-based loss	single	/

- Comprehensive experiments validate the effectiveness and superiority of MRAG, demonstrating that MRAG establishes a new state-of-the-art benchmark.

II. RELATED WORKS

In this section, we summarize advancements in image hiding, including spatial image hiding and JPEG image hiding.

A. Spatial Image Hiding

The cover type that the spatial image hiding focuses on is lossless spatial RGB images. Seminal work [10] designs an image hiding model containing a secret pre-processing module, a hiding module, and a revealing module. The three modules are built with vanilla convolution without special design. It adopts the commonly used pixel-level L2 loss to constrain visual imperceptibility and secret restorability. The extension work [11] further explores the hiding of multiple

images, employing the original model structure and loss constraint. UDH [12] is a cover-agnostic image hiding model that discards the pre-processing structure and solely encodes secret images. The hiding module adopts a simplified U-Net [24], and the revealing module stacks several vanilla convolutional layers. The loss constraint still adopts the L2 loss. ISN [9] treats embedding and extraction as a pair of inverse problems and utilizes the forward and backward processes of invertible neural networks (INN) to address them. The model is composed of a series of affine coupling layers, each of which uses three 5-layer DenseNet blocks. ISN uses the L2 and L1 loss functions for the forward and backward processes, respectively. To improve visual imperceptibility, HiNet [13] combines INN with the discrete wavelet transform (DWT) and introduces low-frequency wavelet loss. DeepMIH [14] proposes a novel framework for multiple image hiding based on enhanced affine coupling layers. It designs an importance map module to guide the concealing process of the next secret

image. It also introduces additional perceptual loss, importance loss, and stage loss to ensure the quality of the generated images. DenseJIN [15] densely connects the enhanced affine coupling layer and incorporates a noninvertible Unet-based module into the forward-hiding process. It adopts the losses used in HiNet. EFCA-DIH [16] utilizes the Sobel operator and DWT to extract wavelet sub-band features with the edge features. It also introduces the coordinate attention mechanism into the affine coupling layer. An additional edge loss is designed to constrain the edge differences between original images and generated images.

The aforementioned methods primarily focus on capacity, secret restorability, and visual imperceptibility. The following works explore the steganalysis imperceptibility utilizing GAN or adversarial attacks. CAIS [18] devises the global and part discriminators and promotes steganalysis imperceptibility through generative adversarial training. The total loss consists of L1 loss, perceptual loss, and cross-entropy loss based on the discriminators' output. Both USAP [19] and DIH-OAIN [20] utilize adversarial attacks to boost steganalysis imperceptibility. DIH-OAIN can extract image semantics and details with dual-resolution branches and reduces computation complexity by balancing image resolution and model complexity. These adversarial attacks-based methods typically employ a pre-trained steganalyzer and introduce an adversarial loss based on the surrogate steganalyzer's output. RIIS [17] is a robust SIH method that considers various distortions. It introduces a conditional INN to model the distribution of the redundant high-frequency component. A distortion-guided modulation is designed to modulate the parameters for different distortion levels. The fourier loss is used to enhance the visual imperceptibility. Nevertheless, the aforementioned works have common shortcomings. The cover type is always lossless spatial RGB images. Their model design lacks steganalysis domain knowledge, relying solely on single-domain feature extraction and attacks. Additionally, the adversarial loss is typically computed as cross-entropy loss based on the surrogate steganalyzer's output.

B. JPEG Image Hiding

EFDR [22] is the first to research image hiding in color JPEG images, focusing on mining fine-grained DCT representations using the transformer-based affine coupling operator. It adopts the commonly used pixel-level L2 loss to constrain visual imperceptibility and secret restorability. EFDR only evaluates steganalysis imperceptibility experimentally and does not consider it in the model design. Therefore, although EFDR has favorable visual imperceptibility and secret restorability, its steganalysis imperceptibility remains poor due to single-range feature extraction and insufficient loss constraints. Noted that, in this paper, we also perform image hiding in color JPEG images, the ubiquitous lossy image format. The proposed MRAG can address the aforementioned shortcomings within a unified framework, enabling steganalysis imperceptibility in color JPEG image hiding while ensuring secret restorability and visual imperceptibility. This is the first attempt to explore steganalysis imperceptibility in color JPEG image hiding, advancing toward more imperceptible color JPEG image hiding.

Table I systematically summarizes the aforementioned image hiding methods from three dimensions: cover type, model architecture, and performance metrics.

III. PROPOSED FRAMEWORK

A. Overview

We propose a novel adversarial stego generation framework, MRAG, for color JPEG image hiding. The goal of the MRAG is to meet the requirements of secret restorability, visual imperceptibility, and steganalysis imperceptibility. The pipeline of MRAG consists of four key components: the invertible local-range branch, the invertible global-range branch, the adaptive fusion module, and the surrogate steganalyzer, as shown in Fig. 2. We construct the proposed MRAG utilizing INN, the mainstream learning paradigm for image hiding, to ensure reversibility as much as possible. The other image conversion operations and network structures are also reversible and symmetric. Therefore, we can implement hiding and revealing utilizing MRAG's forward and backward processes.

The inputs of MRAG's forward-hiding process are the color JPEG cover image C_i and the spatial RGB secret image S_i . Conversion operations contain the coarse-grained frequency decomposition (CFD), the fine-grained frequency decomposition (FFD), and the inverse fine-grained frequency decomposition (IFFD). The inputs are converted into coarse-grained and fine-grained frequency maps through CFD and CFD $\rightarrow$ FFD, respectively, where $\rightarrow$ denotes sequential processing, i.e., the inputs are first processed by CFD and then by FFD. According to the shape characteristic of the obtained frequency maps, they are fed into the local and global branches' forward processes, respectively. Specifically, the shapes of coarse-grained and fine-grained frequency maps are $(B, 3 \times 2, H, W)$ and $(B, 8 \times 8 \times 3 \times 2, \frac{H}{8}, \frac{W}{8})$, respectively, denoting (*batch size, channel number, map height, map width*). H and W represent the image size. Therefore, the shape of fine-grained frequency maps is better suited for the patch-wise architecture of the transformer. Then, the global branch's output passes through the IFFD and is concatenated with the local branch's output. Afterward, the concatenation is fed into the adaptive fusion module's forward process to obtain the JPEG stego image S_o and the redundant information R_o , an all-zero matrix. MRAG's backward-revealing process takes the color JPEG stego image S_o and R_o as input to reveal the secret image. For inducing the surrogate steganalyzer to misclassify the cover C_i and stego S_o , we conduct the features angle-norm disentanglement and pull together the angles between the classified features of cover $\mathbf{S}^2(C_i)$ and the classified features of stego $\mathbf{S}^2(S_o)$ while keeping their norm values changeless.

B. MRAG Model

Backbone structure. Given that INN, the mainstream learning paradigm for image hiding, can preserve as much detail as possible due to its theoretical information lossless characteristic [27]–[29], we utilize it to construct our MRAG to the greatest extent. As shown in Fig. 2, key components of MRAG, including the invertible local-range branch, the invertible global-range branch, and the adaptive fusion module,

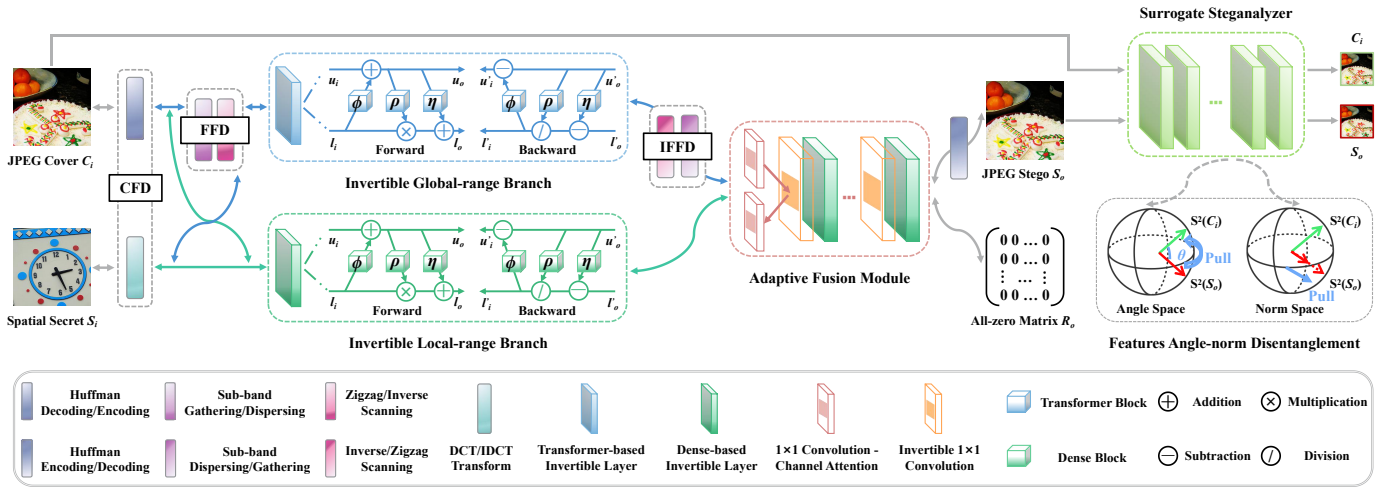


Fig. 2. Overall architecture. To reduce information loss, we utilize INN, the mainstream learning paradigm for image hiding, to construct the proposed MRAG to the greatest extent. The other image conversion operations and network structures are also reversible and symmetric. In the MRAG's forward-hiding process, the spatial RGB secret image S_i is hidden in the color JPEG cover image C_i through the invertible local-range branch, the invertible global-range branch, and the adaptive fusion module to obtain the JPEG stego image S_o and the redundant information R_o , an all-zero matrix. In the MRAG's backward-revealing process, S_o and R_o are fed to reveal the secret image. Features angle-norm disentanglement disentangles the classified features learned by the surrogate steganalyzer. For effectively fooling the surrogate steganalyzer, we pull together the angles between the classified features of cover $S^2(C_i)$ and the classified features of stego $S^2(S_o)$ while keeping their norm values changeless.

employ invertible neural networks. For the affine coupling layer, the basic unit of INN, the input needs to be divided into two parts, u_i and l_i , along the channel axis. Following the reference [30], we use an additive affine transformation for the upper part, u_i , and employ an enhanced affine transformation for the lower part, l_i . To be specific, the forward process of the affine coupling layer can be formulated as

$$\begin{aligned} u_o &= u_i + \phi(l_i) \\ l_o &= l_i \odot \exp(\rho(u_o)) + \eta(u_o) \end{aligned} \quad (1)$$

In Eq. (1), $\phi(\cdot)$, $\rho(\cdot)$, and $\eta(\cdot)$ can be arbitrary network structures. $\odot$ and $\exp(\cdot)$ are the Hadamard product and Exponential function, respectively. Note that $\exp(\cdot)$ is omitted in Fig. 2. Correspondingly, the inverse process is

$$\begin{aligned} l'_i &= (l'_o - \eta(u'_o)) \odot \exp(-\rho(u'_o)) \\ u'_i &= u'_o - \phi(l'_i) \end{aligned} \quad (2)$$

In the following, we will provide a detailed description of the four key components of MRAG.

Invertible local-range branch. As we know, the convolution operator performs excellently in image-related tasks and has nice local neighbor reception characteristics [31]. Here, in the invertible local-range branch, $\phi(\cdot)$, $\rho(\cdot)$, and $\eta(\cdot)$ are implemented by a classical 5-layer dense block with a 3×3 convolution kernel [32]. In the dense block, the output feature maps of each layer are passed to subsequent layers, greatly enhancing representation sharing and learning. For the invertible local-range branch, the input is coarse-grained frequency maps obtained only through CFD. CFD includes the Huffman decoding of the color JPEG cover image and the DCT transformation of the spatial RGB secret image.

Invertible global-range branch. To further enhance representation and global dependency modeling, $\phi(\cdot)$, $\rho(\cdot)$, and $\eta(\cdot)$ in the invertible global-range branch adopt a standard 1-layer

Pre-LN transformer [33]. The Pre-LN transformer applies the layer normalization before the multi-head self-attention and the multi-layer perceptron for a more stable gradient distribution. For the invertible global-range branch, the input is the fine-grained frequency maps obtained through CFD $\rightarrow$ FFD. FFD includes the sub-band gathering and the zigzag scanning. The sub-band gathering rearranges coarse-grained frequency maps to place the frequency components of the same sub-band on the same channel, making frequency components more compact. The zigzag scanning further rearranges them along the channel axis, grouping similar sub-bands and exploiting the structural information in the JPEG transformation.

Adaptive fusion module. In addition, to enhance the interaction between local and global representations extracted by the aforementioned two branches, the proposed adaptive fusion module consists of a pair of 1×1 convolutions with channel attention and a series of affine coupling layers with an invertible 1×1 convolution. $\phi(\cdot)$, $\rho(\cdot)$, and $\eta(\cdot)$ in the adaptive fusion module also employ the classical 5-layer dense block. The 1×1 convolution with channel attention, designed in the reference [34], is used to extract the most informative frequency representations adaptively. Adding an invertible 1×1 convolution before each affine coupling layer can increase the dependencies between representations, enabling more effective information fusion. For the adaptive fusion module, the input is the concatenation of representations extracted from the invertible local-range branch and representations extracted from the invertible global-range branch after undergoing IFFD. IFFD is used for shape alignment.

Surrogate steganalyzer. The surrogate steganalyzer in MRAG can be any well-trained JPEG steganalyzer. Here, we select three open-source steganalyzers: UCNet [35], EWNNet [36], and DBS2Net [37]. UCNet is the sole steganalyzer tailored for color JPEG images. The others are straightforwardly

based on existing steganalyzers for gray-scale JPEG images or some well-known computer vision models. EWNNet and DBS2Net are gray-scale JPEG steganalyzers. Here, following the setting in the reference [35], we adjust the number of input channels in the first convolution layer from 1 to 3, enabling EWNNet and DBS2Net to detect color images. Next, we provide a detailed description of three surrogate steganalyzers based on the papers and open-source codes.

UCNet includes preprocessing, convolutional, and classification modules. The preprocessing module includes three steps: color channel separation, high-pass filters & truncated linear unit, and feature map concatenation. The convolutional module consists of five convolutional groups. Groups 1 and 5 contain two identical layers, each of which includes a 3×3 convolution, batch normalization (BN), and a ReLU activation. Groups 2 and 4 contain a backbone layer and a residual connection structure. The backbone layer consists of a 3×3 convolution, BN, and ReLU, followed by a 3×3 convolution, BN, average pooling, and ReLU. The residual connection structure consists of a 1×1 convolution and BN. Group 3 also contains a backbone layer and a residual connection structure. The backbone layer begins and ends with a 1×1 convolution, BN, and ReLU, while the middle part consists of 3×3 grouped convolution, BN, and ReLU. The residual connection structure contains a 3×3 convolution and BN. The classification module utilizes global average pooling and a fully connected layer.

EWNNet contains four parts: preprocessing, feature extraction, deconvolution, and element-wise classification. The preprocessing part includes a 3×3 convolution and BN. The feature extraction part consists of nine convolutional blocks. It is divided into layer2...layer10. Two kinds of convolutional blocks with a residual connection structure are alternately applied to build the feature extraction module. The backbone layer of the convolutional block is two 3×3 convolution, followed by BN and ReLU. The residual connection structure of one convolutional block includes a 3×3 convolution and BN, while the other does not involve any convolution operation. The deconvolution phase contains three 3×3 deconvolution layers. It is divided into layer11...layer13. The element-wise classification part adopts softmax and mean operations.

DBS2Net is composed of three stages: preprocessing, feature extraction, and classification. The preprocessing stage comprises a 5×5 convolution initialized with SRM filters, the dilated blind-spot network (DBS), and the transition block. DBS involves a 3×3 centrally masked convolution and six dilated convolution residual (DCR) blocks. The transition block consists of two branches: a separable convolution branch and a conventional convolution branch. The feature extraction stage contains six convolutional layers. It is divided into layer5...layer10. Each layer includes a 3×3 convolutional, BN, and ReLU. An average pooling layer is applied after every two layers. The classification stage employs the global average pooling and the fully connected layer.

C. Loss Function

Features angle-norm disentanglement loss. References [25], [26] unveil that the features before the last fully connected layer of the classifier, abbreviated as classified features,

can naturally disentangle into angle and norm distances related to inter-class semantic differences and intra-class confidence, respectively. Specifically, they train a plain CNN classifier on the MNIST dataset. Then set the output dimension of classified features to 2 and visualize these features. Three key observations emerge from the visualization. 1) A well-trained classifier can effectively cluster features belonging to the same category. 2) Features angle accounts for semantic/label differences. 3) Features norm reflects intra-class variations. Inspired by these facts, we introduce features angle-norm disentanglement to compute adversarial loss, which pulls together the angles between the classified features of cover and stego while keeping their norm values changeless.

For fooling the surrogate steganalyzer, we focus on pulling together the angles between the classified features of cover $\mathbf{S}^2(C_i)$ and the classified features of stego $\mathbf{S}^2(S_o)$. Additionally, to minimize the adversarial perturbations, we keep the norm values of $\mathbf{S}^2(C_i)$ and $\mathbf{S}^2(S_o)$ changeless. Thus, we can define the features angle-norm disentanglement loss $\mathcal{L}_{an}$ as follows

$$\begin{aligned}\mathcal{L}_{an} &= -\mathcal{L}_a + \mathcal{L}_n \\ \mathcal{L}_a &= CS(\mathbf{S}^2(C_i), \mathbf{S}^2(S_o)) \\ \mathcal{L}_n &= MSE(\mathbf{S}^2(C_i), \mathbf{S}^2(S_o))\end{aligned}\quad (3)$$

where CS indicates the cosine similarity. CS measures the similarity between two feature vectors by the cosine value of the angle between them. A larger value indicates a smaller angle, hence a higher similarity. MSE is the mean squared error. $\mathcal{L}_a$ and $\mathcal{L}_n$ denote the angle loss and norm loss of the classified features, respectively.

Total loss. The total loss $\mathcal{L}_{total}$ consists of three parts: the hiding loss $\mathcal{L}_{hi}$ to ensure visual imperceptibility, the revealing loss $\mathcal{L}_{re}$ to ensure secret restorability, and the features angle-norm disentanglement loss $\mathcal{L}_{an}$ to guarantee steganalysis imperceptibility. $\mathcal{L}_{total}$ is the sum of these three losses, expressed as follows

$$\mathcal{L}_{total} = \mathcal{L}_{hi} + \mathcal{L}_{re} + \mathcal{L}_{an} \quad (4)$$

where $\mathcal{L}_{hi} = MSE(C_i, S_o)$, and $\mathcal{L}_{re} = MSE(S_i, S'_i)$. S'_i represents the actual restored secret image. Noted that we do not set any restrictions on the redundant information R_o and the recovered cover JPEG image, because these two variables are less important in the whole hiding and revealing tasks. The loss weight is set to 1 by default.

IV. EXPERIMENTS

In this part, experimental settings are described in Subsection A. Numerical and visual comparisons of the proposed MRAG with baselines in terms of secret restorability, visual imperceptibility, and steganalysis imperceptibility are reported in Subsection B. Ablation studies about the number of the affine coupling layer, multi-range branches, features angle-norm disentanglement loss and conventional cross-entropy loss, and coarse/fine-grained maps feeding into local/global branches are conducted in Subsection C. Supplementary discussions about the computational overhead and the features angle-norm disentanglement loss based on different layer indexes' features are provided in Subsection D.

TABLE II
NUMERICAL COMPARISONS OF SECRET RESTORABILITY. MRAG SIGNIFICANTLY OUTPERFORMS EXISTING MODELS. IMAGE SIZE: 128×128 .

Method	COCO				ImageNet				BOSSBase			
	PSNR $\uparrow$	SSIM $\uparrow$	APD $\downarrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	APD $\downarrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	APD $\downarrow$	LPIPS $\downarrow$
ISN+ [9]	19.85	.662	17.52	.344	19.81	.643	18.98	.361	22.55	.731	14.60	.345
RIIS [17]	24.98	.798	9.28	.169	24.97	.781	9.80	.178	29.20	.864	5.84	.143
EFDR [22]	33.95	.918	4.31	.028	34.05	.919	4.37	.027	36.43	.921	3.22	.032
MRAG (UCNet)	40.11	.983	1.88	.004	39.85	.980	2.04	.004	41.54	.975	1.66	.005
MRAG (EWNNet)	41.67	.988	1.62	.001	41.20	.985	1.80	.001	42.32	.978	1.55	.002
MRAG (DBS2Net)	37.14	.978	2.78	.006	37.00	.973	2.90	.005	39.49	.969	2.15	.007

TABLE III
NUMERICAL COMPARISONS OF SECRET RESTORABILITY. MRAG STILL MAINTAINS EXCELLENT SECRET RESTORABILITY. IMAGE SIZE: 256×256 .

Method	COCO				ImageNet				BOSSBase			
	PSNR $\uparrow$	SSIM $\uparrow$	APD $\downarrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	APD $\downarrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	APD $\downarrow$	LPIPS $\downarrow$
EFDR [22]	27.59	.788	8.07	.164	27.80	.790	8.04	.163	32.43	.859	4.77	.143
MRAG (UCNet)	41.02	.987	1.63	.002	40.70	.983	1.80	.003	42.03	.975	1.56	.003
MRAG (EWNNet)	41.28	.987	1.64	.002	40.89	.984	1.81	.002	42.24	.976	1.54	.003
MRAG (DBS2Net)	38.10	.979	2.44	.008	37.88	.976	2.58	.008	40.36	.970	1.95	.010

A. Experimental Settings

Datasets. We employ three image datasets in our implementation, including COCO [38], ImageNet [39], which are widely used in various deep learning tasks, and a standard steganographic dataset, BOSSBase [40]. We utilize 5000, 1000, and 2000 cover-secret pairs from COCO for training, validation, and testing, respectively. 2000 cover-secret pairs, each from ImageNet and BOSSBase, are used only for testing. Additional 10000 cover-stego pairs, randomly selected from three datasets, are used to train the steganalyzer. The stego images are generated by HRJS [41], a bit-level JPEG steganographic method based on deep learning. All images are center-cropped to the common image sizes of 128×128 and 256×256 . The compression quality factor adopted by the cover is 75, the most commonly used in JPEG steganography.

Baselines. Notably, existing hiding works are almost SIH methods. EFDR [22] is the sole JIH method customized for image hiding using color JPEG images as covers with the undeniable comparison value. Additionally, we use two representative SIH methods, ISN [9], and RIIS [17], as our baselines. ISN and RIIS also utilize the mainstream learning paradigm, INN. For SIH methods, when hiding the secret image into the JPEG cover image, to keep the type of the stego image consistent with the cover image and extract the hidden image from the JPEG stego image, it is necessary to suffer from the process of decompression (for the hiding network), compression (from spatial stego image to JPEG stego image) and decompression (for the revealing network). RIIS is a robust SIH method that considers various distortions including JPEG compression. Moreover, during the training phase of ISN, we also add an additional JPEG compression distortion layer, referred to as ISN+ for distinction.

Evaluation metrics. To measure the secret restorability and visual imperceptibility, we adopt peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM) [42], av-

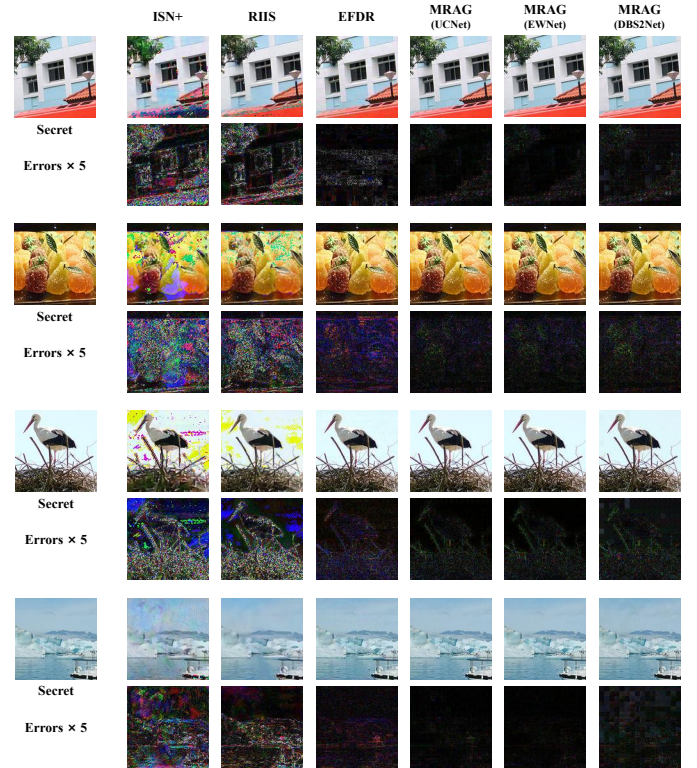


Fig. 3. Visual comparisons of secret restorability. Secret images revealed by MRAG are with minimal visual artifacts. Image size: 128×128 .

erage pixel discrepancy (APD), and learned perceptual image patch similarity (LPIPS) [43]. Higher PSNR/SSIM values and lower APD/LPIPS indicate higher image quality. For steganalysis imperceptibility, we utilize the detection accuracy rate as the metric. The detection accuracy rate is closer to 50%, i.e., a random guess, indicating superior performance. A_U, A_E, and A_D denote the detection accuracy rate of UCNet,

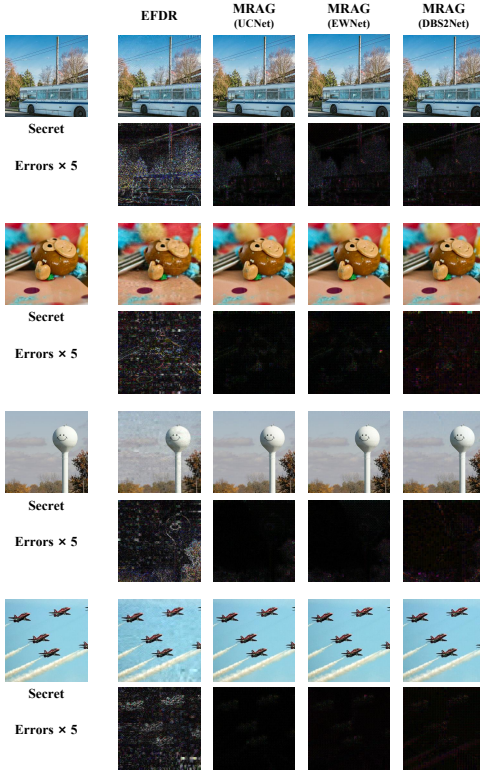


Fig. 4. Visual comparisons of secret restorability. Secret images revealed by MRAG still are with minimal visual artifacts. Image size: 256×256 .

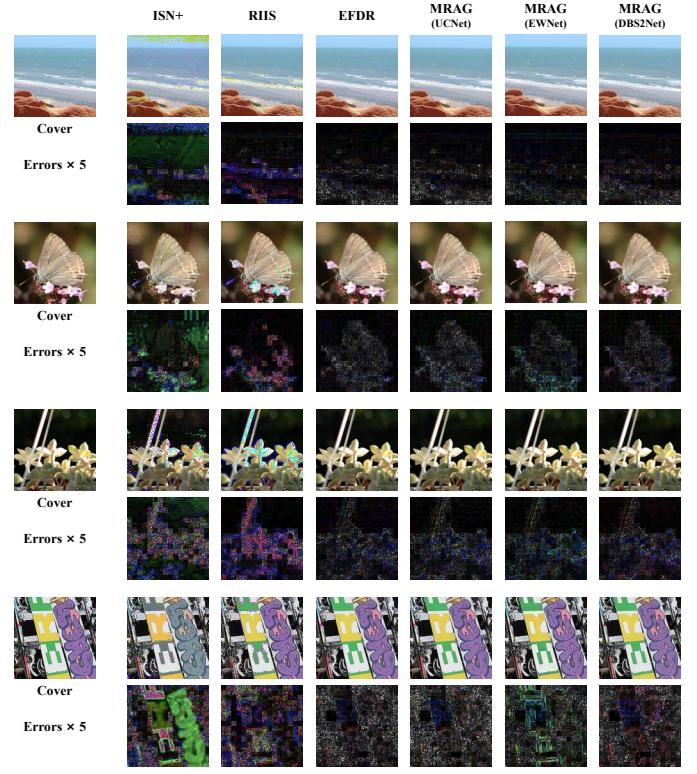


Fig. 5. Visual comparisons of visual imperceptibility. Stego images generated by MRAG are visually friendly. Image size: 128×128 .

EWNNet, and DBS2Net, respectively. Given two images X and Y , the evaluation metrics can be expressed as follows.

- PSNR assesses the clarity and signal intensity. It is defined as

$$PSNR = 10 \log_{10} \left(\frac{MAX^2}{MSE} \right) \quad (5)$$

where MAX represents the maximum pixel value. MSE is the mean squared error.

- SSIM evaluates structural similarity. It is computed as

$$SSIM = \frac{(2\mu_X\mu_Y + c_1)(2\sigma_{XY} + c_2)}{(\mu_X^2 + \mu_Y^2 + c_1)(\sigma_X^2 + \sigma_Y^2 + c_2)} \quad (6)$$

where μ , σ , and σ^2 denote mean, covariance, and variance, respectively. c_1 and c_2 are two constants.

- APD measures the average pixel-wise discrepancy. It is expressed as

$$APD = \frac{1}{N} \sum_{i=1}^N |X - Y| \quad (7)$$

where N is the number of pixels.

- LPIPS computes perceptual similarity using deep feature representations. It is expressed as

$$LPIPS = \sum_l \frac{1}{H_l W_l} \sum_{h,w} \|\bar{\omega}_l \odot (\psi_l(X) - \psi_l(Y))\|_2^2 \quad (8)$$

where $\psi_l(\cdot)$ denotes deep features extracted from layer l , and $\bar{\omega}_l$ represents learned channel-wise weights.

- The detection accuracy rate is defined as

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (9)$$

where TP , TN , FP , and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively.

Implementation details. Before training MRAG, the surrogate steganalyzer is pre-trained for 50 epochs on 5000 mixed cover-stego pairs following default settings. Subsequently, the pre-trained surrogate steganalyzer's weights are frozen, and we train MRAG for 200 epochs on 5000 cover-secret pairs from COCO. Noted that the steganalyzer used for the steganalysis imperceptibility evaluation is independent of MRAG and is trained on another 5000 mixed cover-stego pairs. We adopt the Adam optimizer with $\text{betas} = (0.5, 0.999)$, $\text{eps} = 1e^{-6}$, and $\text{weight_decay} = 5e^{-4}$, and set the learning rate to $5e^{-4}$. The mini-batch size is 4, containing two randomly selected cover-secret pairs. All experiments are implemented with PyTorch on the NVIDIA GeForce RTX 4090 GPU.

B. Comparison Results

To verify the performance of MRAG in terms of secret restorability, visual imperceptibility, and steganalysis imperceptibility, we conduct extensive experiments on three datasets and three open-source JPEG steganalyzers.

Secret restorability. Table II reports the numerical comparisons of MRAG and baselines when the image size is 128×128 . It can be observed that MRAG significantly

TABLE IV
NUMERICAL COMPARISONS OF VISUAL IMPERCEPTIBILITY. MRAG AND EFDR HAVE COMPARABLE PERFORMANCE. IMAGE SIZE: 128×128 .

Method	COCO				ImageNet				BOSSBase			
	PSNR $\uparrow$	SSIM $\uparrow$	APD $\downarrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	APD $\downarrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	APD $\downarrow$	LPIPS $\downarrow$
ISN+ [9]	24.09	.828	9.96	.208	24.08	.823	10.66	.213	27.12	.830	7.80	.239
RIIS [17]	28.25	.897	5.63	.098	28.56	.895	5.86	.104	32.76	.921	3.93	.102
EFDR [22]	32.22	.927	4.71	.025	32.04	.924	4.93	.027	36.01	.944	3.12	.027
MRAG (UCNet)	31.68	.920	5.01	.028	31.53	.916	5.24	.031	35.60	.940	3.27	.031
MRAG (EWNNet)	32.78	.936	4.39	.023	32.62	.934	4.58	.025	36.37	.950	2.98	.026
MRAG (DBS2Net)	32.35	.930	4.65	.022	32.16	.927	4.86	.024	35.92	.945	3.16	.026

TABLE V
NUMERICAL COMPARISONS OF VISUAL IMPERCEPTIBILITY. MRAG ARE WITH OPTIMAL PERFORMANCE. IMAGE SIZE: 256×256 .

Method	COCO				ImageNet				BOSSBase			
	PSNR $\uparrow$	SSIM $\uparrow$	APD $\downarrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	APD $\downarrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	APD $\downarrow$	LPIPS $\downarrow$
EFDR [22]	28.16	.833	7.06	.122	28.24	.831	7.12	.123	32.23	.878	4.49	.133
MRAG (UCNet)	32.18	.929	4.56	.029	32.19	.927	4.65	.030	36.09	.947	2.99	.033
MRAG (EWNNet)	32.87	.939	4.17	.026	32.89	.938	4.26	.028	36.63	.954	2.75	.031
MRAG (DBS2Net)	32.07	.928	4.61	.027	32.07	.925	4.71	.030	35.91	.944	3.04	.033

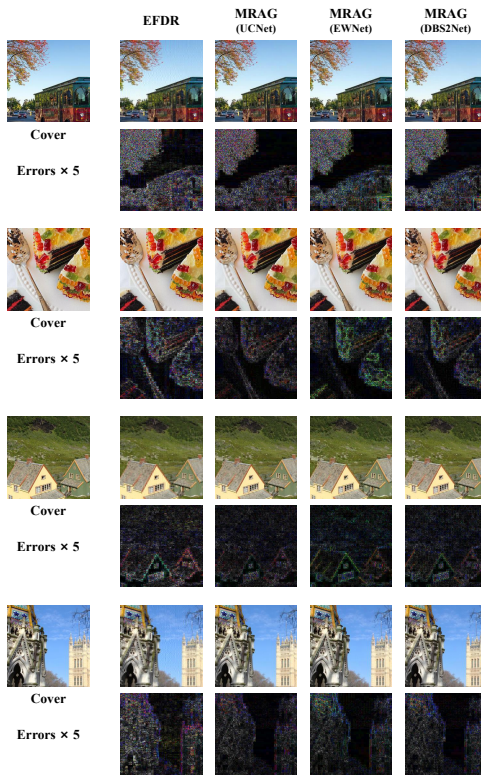


Fig. 6. Visual comparisons of visual imperceptibility. Stego images generated by MRAG still maintain visually friendly. Image size: 256×256 .

outperforms these baselines in terms of secret restorability. Compared with EFDR, MRAG can supply at least 2.95 dB PSNR advancement. Similar performance improvements can also be observed in SSIM, APD, and LPIPS. Corresponding visual comparisons are shown in Fig. 3. We can observe that the secret images revealed by MRAG are visually pleasing with minimal visual artifacts. In contrast, the SIH methods ISN+

and RIIS yield visible color distortion, blocking, blurring, and banding artifacts in the revealed secret images. We further conduct experiments on the image size of 256×256 . Table III and Fig. 4 present the numerical comparisons and visual comparisons, respectively. We can observe that MRAG still maintains excellent secret restorability. In the experiments with an image size of 256×256 , we omitted the comparison with the SIH methods. It is noteworthy that although our model only uses the COCO for training, it still provides excellent results on the other two datasets, which indicates that our model has a favorable generalization ability.

Visual imperceptibility. Table IV and Fig. 5 report the numerical comparisons and visual comparisons of our MRAG with ISN+, RIIS, and EFDR in terms of visual imperceptibility when the image size is 128×128 . We can observe that MRAG and EFDR have comparable performance, while the stegos generated by ISN+ and RIIS still contain many visible artifacts, making it difficult for them to achieve visual imperceptibility. Additionally, Table V and Fig. 6 show the comparisons of MRAG and EFDR on the image size of 256×256 . They also demonstrate that the stego images generated by MRAG have optimal performance.

Steganalysis imperceptibility. Steganalysis imperceptibility is a more rigorous imperceptibility metric, evaluating the ability to resist steganalyzers. It achieves classification by distinguishing the changes in statistical features caused by hiding. Table VI shows the numerical comparisons of our MRAG and EFDR when the image size is 128×128 . We can observe that MRAG significantly outperforms EFDR. Compared to EFDR, MRAG can provide at least 15.2% improvement, with most classification results approaching 50%. Moreover, we visualize the distribution of the classified features learned by three steganalyzers using t-distributed stochastic neighbor embedding (t-SNE) in Fig. 7. We can find that the distribution of covers' classified features (green points) and stegos' classified features

TABLE VI

STEGANALYSIS IMPERCEPTIBILITY COMPARISONS. MOST OF MRAG'S CLASSIFICATION RESULTS APPROACH 50%, I.E., A RANDOM GUESS. MRAG MAINTAINS HIGHER STEGANALYSIS IMPERCEPTIBILITY EVEN ACROSS THE STEGANALYZER. IMAGE SIZE: 128×128 .

Method	COCO				ImageNet				BOSSBase			
	A_U↓	A_E↓	A_D↓	Average↓	A_U↓	A_E↓	A_D↓	Average↓	A_U↓	A_E↓	A_D↓	Average↓
EFDR [22]	92.73	75.68	99.90	89.44	92.85	75.65	99.90	89.47	87.65	73.70	99.98	87.11
MRAG (UCNet)	50.05	51.28	74.53	58.62	50.00	51.68	78.83	60.17	50.00	50.55	63.28	54.61
MRAG (EWNNet)	58.53	60.48	71.60	63.54	60.73	59.25	74.65	64.88	51.20	55.00	59.05	55.08
MRAG (DBS2Net)	53.35	51.88	50.13	51.79	53.50	51.18	50.30	51.66	51.25	50.43	50.13	50.60

TABLE VII

STEGANALYSIS IMPERCEPTIBILITY COMPARISONS. MRAG STILL MAINTAINS HIGHER STEGANALYSIS IMPERCEPTIBILITY. IMAGE SIZE: 256×256 .

Method	COCO				ImageNet				BOSSBase			
	A_U↓	A_E↓	A_D↓	Average↓	A_U↓	A_E↓	A_D↓	Average↓	A_U↓	A_E↓	A_D↓	Average↓
EFDR [22]	99.70	96.38	100.00	98.69	99.80	96.68	99.90	98.79	99.93	99.20	100.00	99.71
MRAG (UCNet)	50.05	59.58	99.60	69.74	50.05	59.53	99.38	69.65	50.00	55.00	97.70	67.57
MRAG (EWNNet)	71.00	56.20	99.40	75.53	72.05	56.20	99.00	75.75	53.63	51.43	95.70	66.92
MRAG (DBS2Net)	67.33	53.13	56.33	58.93	67.70	53.05	58.05	59.60	56.73	50.30	53.40	53.48

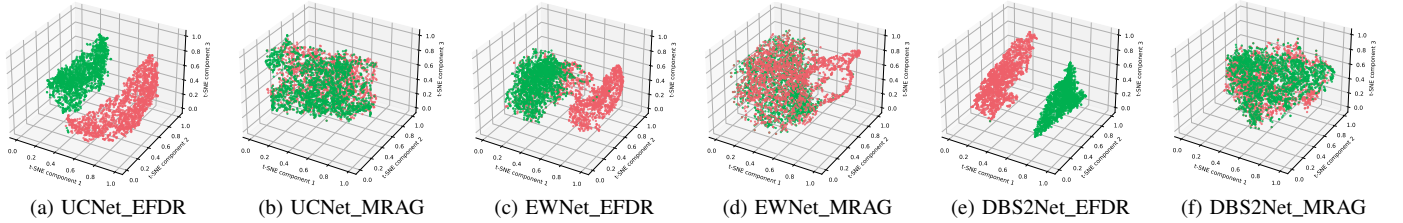


Fig. 7. The t-SNE visualization comparisons of the distribution of the classified features learned by three steganalyzers. Green points denote the distribution of covers' classified features, and red points show the distribution of stegos' classified features. The greater the distinction between two distributions, the easier they are detected by steganalyzers. Image size: 128×128 .

(red points) obtained by EFDR can be well distinguished, while the distribution of the classified features obtained by the proposed MRAG is intermingled and difficult to distinguish. This also confirms the superior steganalysis imperceptibility of MRAG. Additionally, Table VII shows that steganalysis imperceptibility on the image size of 256×256 has decreased somewhat, but it is mostly within an acceptable range. We can also observe that when the surrogate steganalyzer is DBS2Net, steganalysis imperceptibility is the best. This is because DBS2Net has a relatively strong detection ability, making the generated stego more adversarial.

C. Ablation Study

Here, we conducted a series of ablation experiments, including ablation of the number of the affine coupling layer, ablation of multi-range branches, ablation of features angle-norm disentanglement loss and conventional cross-entropy loss, and ablation of coarse/fine-grained maps feeding into local/global branches. The image size used is 128×128 , the surrogate steganalyzer in MRAG is UCNet, and the image test set is COCO, unless otherwise specified.

Ablation of the number of the affine coupling layer. For the convenience of discussion, we fix the number of the affine coupling layer in the invertible local-range branch and the invertible global-range branch to 1, and focus on investigating

the impact of the number of the affine coupling layer in the adaptive fusion module on performance. As reported in Table VIII, we can observe that the steganalysis imperceptibility performance, the most concerned metric, is optimal when the number of layers is 3. Thus, in our implementation, the number of the affine coupling layer in the adaptive fusion module defaults to 3 unless otherwise specified.

Ablation of multi-range branches. Also, we verify the performance effects of the invertible local-range branch and the invertible global-range branch, as shown in Table IX. It can be observed that compared to any single-range branch, the designed multi-range branches can significantly boost the performance of secret restorability and steganalysis imperceptibility while keeping visual imperceptibility.

Ablation of adversarial loss. To further validate the advantages of the proposed features angle-norm disentanglement loss, we compare it with the conventional cross-entropy loss. It can be observed that MRAG demonstrates superior steganalysis and visual imperceptibility with comparable secret restorability, as shown in Table X.

Ablation of coarse/fine-grained maps. We mentioned that according to the shape characteristic of the obtained frequency maps, they are fed into the local and global branches, respectively. The shapes of coarse-grained and fine-grained frequency maps are $(B, 3 \times 2, H, W)$ and $(B, 8 \times 8 \times 3 \times 2, \frac{H}{8}, \frac{W}{8})$

TABLE VIII
ABLATION FOR THE IMPACT OF THE NUMBER OF THE AFFINE COUPLING LAYER IN THE ADAPTIVE FUSION MODULE ON PERFORMANCE.

The Number of Layer	Secret Restorability		Visual Imperceptibility		Steganalysis Imperceptibility			
	PSNR↑	SSIM↑	PSNR↑	SSIM↑	A_U↓	A_E↓	A_D↓	Average↓
1	41.59	.987	33.21	.941	50.05	52.48	75.88	59.47
2	40.56	.984	32.60	.934	50.08	52.85	98.90	67.28
3	40.11	.983	31.68	.920	50.05	51.28	74.53	58.62
4	39.72	.983	31.80	.923	50.05	64.23	80.30	64.86

TABLE IX
ABLATION OF THE INVERTIBLE LOCAL-RANGE BRANCH AND THE INVERTIBLE GLOBAL-RANGE BRANCH.

Local	Global	Secret Restorability		Visual Imperceptibility		Steganalysis Imperceptibility			
		PSNR↑	SSIM↑	PSNR↑	SSIM↑	A_U↓	A_E↓	A_D↓	Average↓
✗	✓	26.23	.786	31.95	.924	50.03	51.15	67.75	56.31
✓	✗	36.20	.965	31.27	.908	50.18	51.10	85.13	62.14
✓	✓	40.11	.983	31.68	.920	50.05	51.28	74.53	58.62

TABLE X
ABLATION OF THE ADVERSARIAL LOSS ON SECRET RESTORABILITY, VISUAL IMPERCEPTIBILITY, AND STEGANALYSIS IMPERCEPTIBILITY.

Adversarial Loss	Secret Restorability		Visual Imperceptibility		Steganalysis Imperceptibility			
	PSNR↑	SSIM↑	PSNR↑	SSIM↑	A_U↓	A_E↓	A_D↓	Average↓
cross-entropy	40.54	.985	29.17	.908	50.13	53.33	85.10	62.85
angle-norm	40.11	.983	31.68	.920	50.05	51.28	74.53	58.62

TABLE XI
ABLATION OF COARSE/FINE-GRAINED MAPS ON SECRET RESTORABILITY, VISUAL IMPERCEPTIBILITY, AND STEGANALYSIS IMPERCEPTIBILITY.

Local	Global	Secret Restorability		Visual Imperceptibility		Steganalysis Imperceptibility			
		PSNR↑	SSIM↑	PSNR↑	SSIM↑	A_U↓	A_E↓	A_D↓	Average↓
fine	fine	22.14	.501	32.22	.928	50.00	50.00	79.70	59.90
coarse	coarse	39.76	.981	32.05	.927	50.03	57.65	98.45	68.71
fine	coarse	39.68	.981	31.95	.924	50.10	50.93	83.90	61.64
coarse	fine	40.11	.983	31.68	.920	50.05	51.28	74.53	58.62

respectively. The shape of fine-grained frequency maps is better suited for the patch-wise architecture of the transformer. Here, we also verify the rationality of this allocation. Table XI reports the impact of coarse/fine-grained maps feeding into local/global branches on performance. We can observe that our allocation strategy has the best performance.

D. Discussion

Here, we further discuss the computational overhead. Additionally, we attempt to compute the features angle-norm disentanglement loss based on different layer indexes' features.

Computational overhead. We further discuss the computational overhead by calculating the number of floating point operations (FLOPs), model parameters (Params), and the inference time for a single image (Times). The results in Table XII show that the proposed MRAG has relatively low overhead and is always optimal or sub-optimal.

Features angle-norm disentanglement loss based on intermediate features. In this paper, the adversarial loss,

TABLE XII
COMPARISONS OF COMPUTATIONAL OVERHEAD. PARAMS ($\times 10^4$), FLOPs ($\times 10^8$), TIMES (s).

Method	Params	FLOPs	Times
ISN+ [9]	74.90	244.93	.009
RIIS [17]	1016.38	3325.17	.102
EFDR [22]	16997.99	652.72	.029
MRAG	14.91	299.35	.017

features angle-norm disentanglement loss, is computed based on the surrogate steganalyzer's classified features. Notably, some studies, such as intermediate-level attack (ILA) [44], feature importance-aware attack (FIA) [45], and multi-feature attention attack (MFAA) [46], show that disrupting intermediate features extracted by the surrogate classifier can improve the transferability of adversarial attacks. Here, as an extension, we further attempt to compute the features angle-norm disentanglement loss based on the features from

TABLE XIII

THE TEST LOSS AND THE STEGANALYSIS IMPERCEPTIBILITY BASED ON DIFFERENT LAYER INDEXES' FEATURES IN THE SURROGATE UCNET.

Feature Index	Test Loss	Steganalysis Imperceptibility			
		A_U↓	A_E↓	A_D↓	Average↓
classified features	11.79	50.05	51.28	74.53	58.62
group4's features	20.14	50.78	52.75	76.53	60.02
group3's features	17.43	52.05	54.05	90.10	65.40
group2's features	11.47	51.53	51.10	69.90	57.51
group1's features	32.37	60.75	59.00	92.88	70.88
pre-processed features	98.67	56.88	59.70	82.20	66.26

TABLE XIV

THE TEST LOSS AND THE STEGANALYSIS IMPERCEPTIBILITY BASED ON DIFFERENT LAYER INDEXES' FEATURES IN THE SURROGATE EWNET.

Feature Index	Test Loss	Steganalysis Imperceptibility			
		A_U↓	A_E↓	A_D↓	Average↓
classified features	20.77	58.53	60.48	71.60	63.54
layer12's features	8.51	50.38	50.20	56.73	52.43
layer11's features	106.41	54.15	54.40	83.65	64.07
layer10's features	104.51	53.10	54.65	65.80	57.85
layer9's features	9.15	50.33	50.00	59.28	53.20
layer8's features	9.10	50.08	50.08	72.00	57.38
layer7's features	40.96	50.00	50.00	58.00	52.67
layer6's features	9.87	50.00	50.00	78.33	59.44
layer5's features	43.07	50.00	50.00	53.30	51.10
layer4's features	9.13	50.00	50.03	53.88	51.30
layer3's features	38.10	50.03	50.00	54.53	51.52
layer2's features	9.48	50.08	50.03	89.40	63.17
pre-processed features	15.28	53.63	53.63	71.95	59.73

other layers of the surrogate steganalyzer. The experimental results are shown in Tables XIII-XV. Blue-background results are obtained by computing the features angle-norm disentanglement loss based on the classified features, while green-background results represent the optimal performance using the intermediate features from other layers. The division of the network layer in the surrogate steganalyzer is based on its paper and source code. Both reduced test loss and decreased detection accuracy indicate improved performance. This extension validates the effectiveness of computing features angle-norm disentanglement loss using intermediate features.

TABLE XV

THE TEST LOSS AND THE STEGANALYSIS IMPERCEPTIBILITY BASED ON DIFFERENT LAYER INDEXES' FEATURES IN THE SURROGATE DBS2NET.

Feature Index	Test Loss	Steganalysis Imperceptibility			
		A_U↓	A_E↓	A_D↓	Average↓
classified features	26.34	53.35	51.88	50.13	51.79
layer9's features	48.45	50.05	53.20	50.00	51.08
layer8's features	22.86	50.88	55.18	50.03	52.03
layer7's features	12.14	50.13	50.00	50.00	50.04
layer6's features	44.11	61.43	57.88	50.10	56.47
layer5's features	16.05	52.80	52.70	50.08	51.86
pre-processed features	14.08	50.08	50.63	50.00	50.23

V. CONCLUSION

This work is the first attempt to explore steganalysis imperceptibility in color JPEG image hiding, advancing toward more imperceptible color JPEG image hiding. It designs a multi-range representations-driven adversarial stego generation framework called MRAG for color JPEG image hiding, enabling steganalysis imperceptibility while ensuring secret restorability and visual imperceptibility. MRAG comprises four key components: the invertible local-range branch, the invertible global-range branch, the adaptive fusion module, and the surrogate steganalyzer. Meanwhile, a loss based on the surrogate steganalyzer's classified features, termed the features angle-norm disentanglement loss, is incorporated to launch multi-range representations-driven feature-level adversarial attacks. Moreover, coarse-grained and fine-grained frequency decomposition operations are devised to introduce multi-grained frequency information. Comprehensive experiments verify the effectiveness and superiority of MRAG. We believe the success of MRAG can further promote the development of color JPEG image hiding and provide some inspiration for further research on steganalysis imperceptibility.

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